Exploring House Prices in Ames, Iowa

Mohammed AlRashed | MLFoundationsCapstoneReport

1. Introduction

In this project the Ames housing dataset was used, which contains data about 2930 residential property sales that is located in Ames, Iowa. Each row record contains a set of features describing the different aspects of the house, along with some information about its sale. I answered some questions using this dataset, such as:(1) What's the impact of different features on the houses' price? (2) Which features are strongly connected to the price of the house?

2. Cleaning Summary

The raw dataset had several problems that needed fixing before analysis:

- **Missing values:**
  After examining the missing values in each column, I have found 6 columns that were mostly null, so I dropped them, they are: (Alley, Mas Vnr Type, Fireplace Qu, Pool QC, Fence Misc Feature)
- **Data types:** As for the datatypes, I found two columns (Bsmt Full Bath, Bsmt Half Bath) that were floats, then I made them integers
- **Outliers:** After using the boxplot chart for the "SalePrice" column, I have found a lot of outliers, but I didn't remove them because they still are real sales that happened.
  After that, I took the values above the 99th percentile of the data, then capped them.
- **Duplicates:** No duplicates have been found.
  After that, I did the three checks: 1 - No nulls, 2- No negatives, 3- Complete number of columns (i only dropped the 6 mentioned above)

Then i made a function named "clean_data" I put all the previous steps to clean a database in that function.

3. Feature Engineering Summary

I made some new columns to get new information, Such as:

- Price_per_SF = SalePrice / LotArea. This variable helps us see how much one sqft costs.
- Area_Of_garage_to_total = This variable helps us see the ratio of the garage area to the total area of the house
- Qual x price = This creates an interactive variable.
- SalePrice_log1p = After seeing that the original data was skewed, I transformed the data to reduce the skew.

I also one-hot encoded two columns, 1- the Neighborhood column (28 categories) and the Sale Condition columns (6 categories) and ordinally encoded the Exter Qual column (1-3 scale). Also, I scaled 2 columns: (Lot Area) (Total Bsmt SF) by standardscaling, so its mean = 0 and std = 1

4. Key Findings

Finding 1 — Overall Quality has the strongest correlation with saleprice. The correlation matrix(fig 1) represents that the column that has the most correlation with the price of the house was the overall quality.

Finding 2 — The years matter. After making a scatter plot that combines the saleprice and overall quality and the year the house was built (Fig. 2), it has come to me that with time, the house prices in Ames, Iowa, have increased, as has the overall quality of the houses

Finding 3 — Zones the house is in significantly affects the price! The grouped boxplots, containing the different zones of Ames, Iowa, and the Saleprice of the houses (fig. 3), has shown that there is a huge difference in house prices depending on the zone they are in.

Finding 4 — How streets matter. A group box plot that has the saleprice and street material as its axises has been made (fig. 4), and it seems that the material of the street the house is on majorly affects its price, showing that the houses on the paved street are significantly more expensive than the houses on gravel.

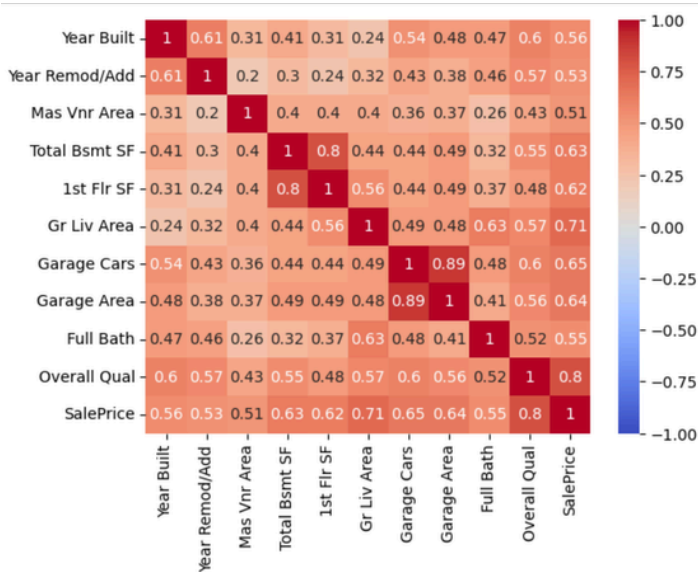

fig 1

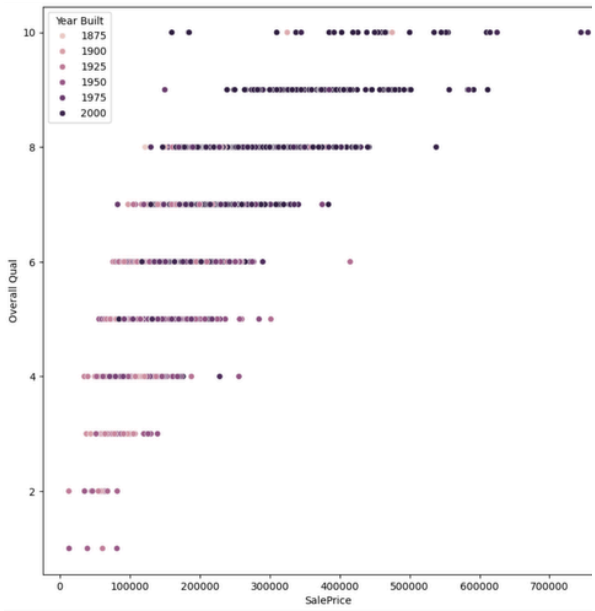

fig 2

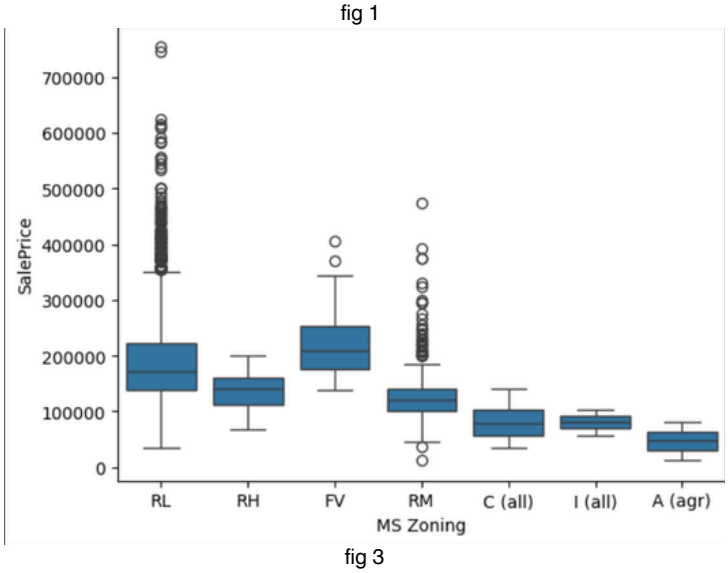

fig 3

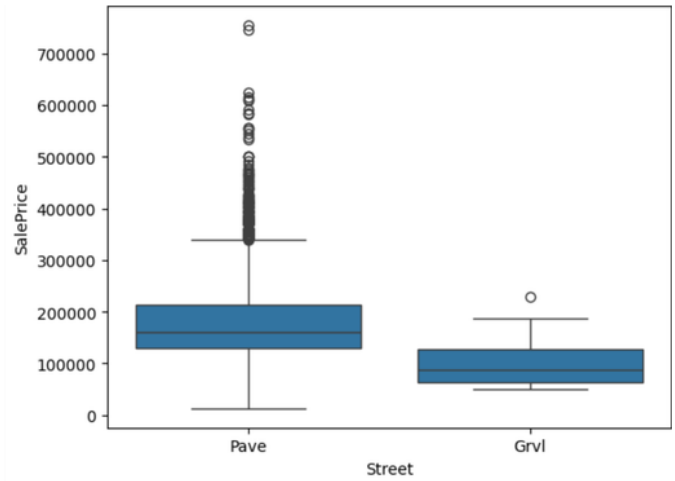

fig 4

5. What I Would Do Next

If I had more time, I would: (1) train a simple linear regression model using some features such as the Lot Area or the total number of bathroom to predict the house prices based on them; (2) see if the number of bathrooms or the total number of rooms affect the house price differently based on the neighborhood; (3) see how the different materials the different parts of the house are made of affect the price; (4) see if the total number of cars the garage holds is associated with the area of the house